Module 9: Inheritance Genetics — Study Questions

1. What organism did Mendel study? Why was it a good choice for genetics experiments?
2. Define the Law of Segregation and the Law of Independent Assortment.
3. Explain the difference between genotype and phenotype, and between homozygous and heterozygous.
4. Cross Tt × Tt. What are the expected genotypic and phenotypic ratios?
5. What is a test cross? Cross Tt × tt and explain what the results tell you.
6. What phenotypic ratio is expected from a dihybrid cross (AaBb × AaBb)?
7. Differentiate between incomplete dominance and codominance. Give an example of each.
8. In snapdragons, red (RR) × white (rr) produces pink (Rr). Cross Rr × Rr. What are the expected phenotypic ratios?
9. Explain the ABO blood type system. What alleles are involved? Which are codominant? Which is recessive?
10. Can a Type A mother (I^Ai) and a Type B father (I^Bi) have a Type O child? Show the Punnett square.
11. What is pleiotropy? How does the sickle cell allele illustrate this concept?
12. Why are males more frequently affected by X-linked recessive disorders (e.g., color blindness, hemophilia)?
13. A carrier mother and an unaffected father have children. What fraction of their sons will be affected? What fraction of their daughters will be carriers?
14. Two unaffected parents have an affected child. Is the trait most likely dominant or recessive? Explain.

15. In a pedigree, a trait appears almost exclusively in males and is never passed from father to son. What inheritance pattern does this suggest?

16. What is a polygenic trait? How does the inheritance of human height differ from pea plant height?

17. Why do polygenic traits show continuous variation instead of distinct categories?

18. A student says: "If I know someone's genotype, I know exactly what they'll look like." Is this accurate? Give a specific example.

19. Explain how Mendel's Law of Segregation connects to the physical process of homolog separation during Meiosis I (link to Module 8).

20. Compare single-gene inheritance (Mendelian) and polygenic inheritance: number of genes, phenotype categories, example traits, and distribution pattern.